

### L3 – 8.1/8.2 Sum/Difference and Product/Quotient of Functions

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#### Part 1: Sum and Difference of Functions

When two functions  $f(x)$  and  $g(x)$  are combined to form the function  $(f + g)(x)$  or  $(f - g)(x)$ , the new function is called the sum or difference of  $f$  and  $g$ .

The graph of  $f + g$  or  $f - g$  can be obtained by adding or subtracting corresponding y-coordinates. This is called the **superposition principle**.

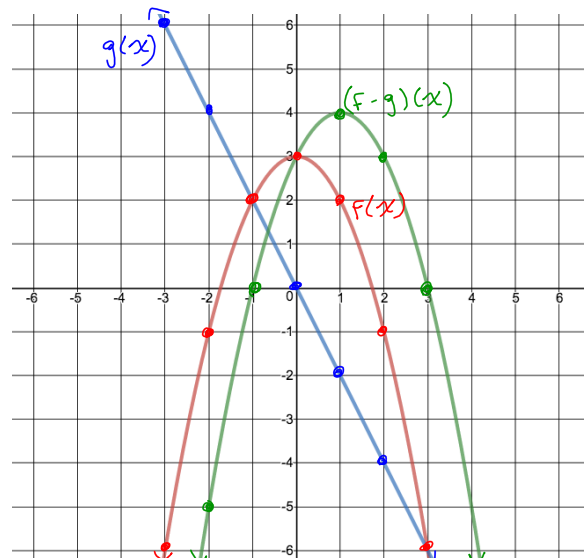
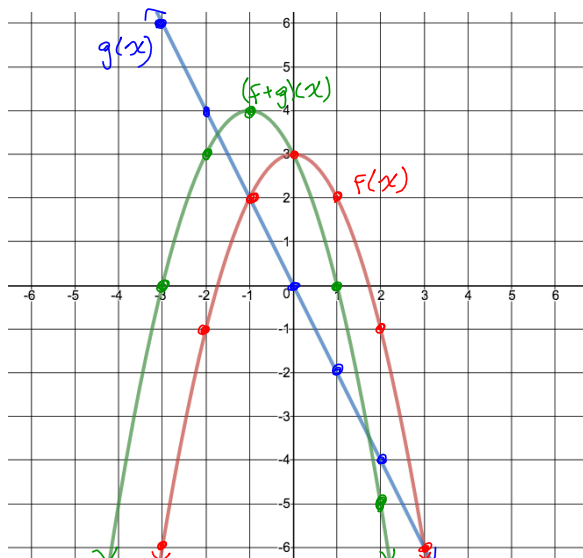
$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

**Example 1:** Given  $f(x) = -x^2 + 3$  and  $g(x) = -2x$  determine the graphs of  $(f + g)(x)$  and  $(f - g)(x)$ .

#### Method 1: Graphically

| $x$ | $f(x)$ | $g(x)$ | $f(x) + g(x)$ | $f(x) - g(x)$ |
|-----|--------|--------|---------------|---------------|
| -3  | -6     | 6      | 0             | -12           |
| -2  | -1     | 4      | 3             | -5            |
| -1  | 2      | 2      | 4             | 0             |
| 0   | 3      | 0      | 3             | 3             |
| 1   | 2      | -2     | 0             | 4             |
| 2   | -1     | -4     | -5            | 3             |
| 3   | -6     | -6     | -12           | 0             |



## Method 2: Algebraically

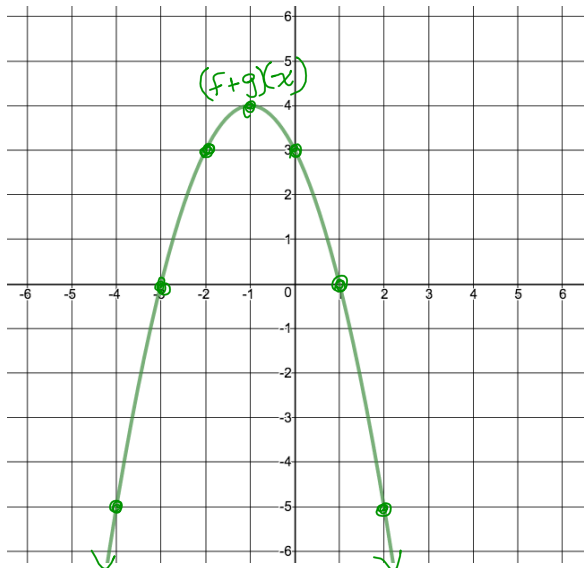
$$\begin{aligned}(f + g)(x) &= f(x) + g(x) \\(f + g)(x) &= (-x^2 + 3) + (-2x) \\(f + g)(x) &= -x^2 - 2x + 3\end{aligned}$$

Complete Square to find Vertex

$$\begin{aligned}(f + g)(x) &= -(x^2 + 2x) + 3 \\(f + g)(x) &= -(x^2 + 2x + 1 - 1) + 3 \\(f + g)(x) &= -(x^2 + 2x + 1) + 1 + 3 \\(f + g)(x) &= -(x + 1)^2 + 4\end{aligned}$$

vertex is (-1, 4)

| $x$ | $(f + g)(x)$ |
|-----|--------------|
| -4  | -5           |
| -3  | 0            |
| -2  | 3            |
| -1  | 4            |
| 0   | 3            |
| 1   | 0            |
| 2   | -5           |



$$(f + g)(x)$$

$$D: \{X \in \mathbb{R}\}$$

$$R: \{Y \in \mathbb{R} | y \leq 4\}$$

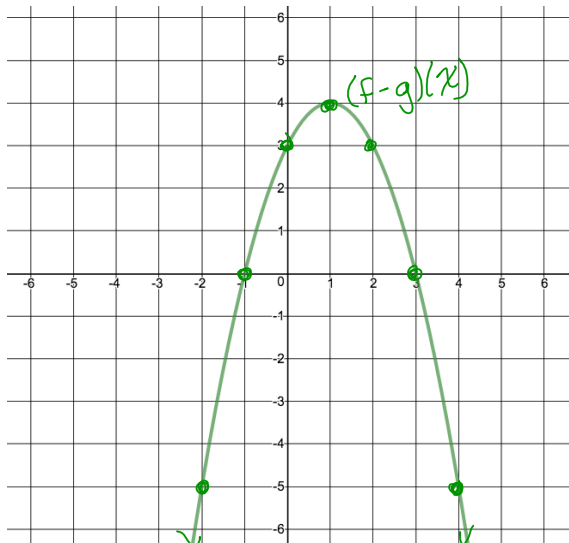
$$\begin{aligned}(f - g)(x) &= f(x) - g(x) \\(f - g)(x) &= (-x^2 + 3) - (-2x) \\(f - g)(x) &= -x^2 + 2x + 3\end{aligned}$$

Complete Square to find Vertex

$$\begin{aligned}(f - g)(x) &= -(x^2 - 2x) + 3 \\(f - g)(x) &= -(x^2 - 2x + 1 - 1) + 3 \\(f - g)(x) &= -(x^2 - 2x + 1) + 1 + 3 \\(f - g)(x) &= -(x - 1)^2 + 4\end{aligned}$$

vertex is (1, 4)

| $x$ | $(f - g)(x)$ |
|-----|--------------|
| -2  | -5           |
| -1  | 0            |
| 0   | 3            |
| 1   | 4            |
| 2   | 3            |
| 3   | 0            |
| 4   | -5           |



$$(f - g)(x)$$

$$D: \{X \in \mathbb{R}\}$$

$$R: \{Y \in \mathbb{R} | y \leq 4\}$$

**Note:** The domain of the sum or difference of functions is the intersection of the domains of  $f$  and  $g$

## Part 2: Product and Quotient of Functions

When two functions  $f(x)$  and  $g(x)$  are combined to form the function  $(f \cdot g)(x)$  or  $(f \div g)(x)$ , the new function is called the product or quotient of  $f$  and  $g$ .

The graph of  $f \cdot g$  or  $f \div g$  can be obtained by multiplying or dividing corresponding  $y$ -coordinates.

$$(f \times g)(x) = f(x) \times g(x)$$

$$(f \div g)(x) = f(x) \div g(x)$$

**Example 2:** Let  $f(x) = x + 3$  and  $g(x) = x^2 + 8x + 15$ . Determine an equation and graph for

**a)**  $(f \times g)(x)$

$$(f \times g)(x) = f(x)g(x)$$

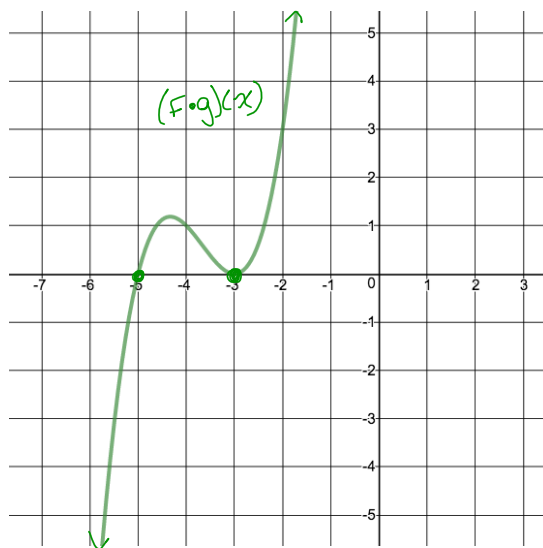
$$(f \times g)(x) = (x + 3)(x^2 + 8x + 15)$$

$$(f \times g)(x) = (x + 3)(x + 3)(x + 5)$$

$$(f \times g)(x) = (x + 3)^2(x + 5)$$

$x$  –intercepts at -3 (order 2) and -5 (order 1)

Extends from Q1 to Q3



b)  $(f \div g)(x)$

$$(f \div g)(x) = \frac{f(x)}{g(x)}$$

$$(f \div g)(x) = \frac{x+3}{(x+3)(x+5)}$$

$$(f \div g)(x) = \frac{1}{x+5}; x \neq -5, -3$$

VA:  $x = -5$

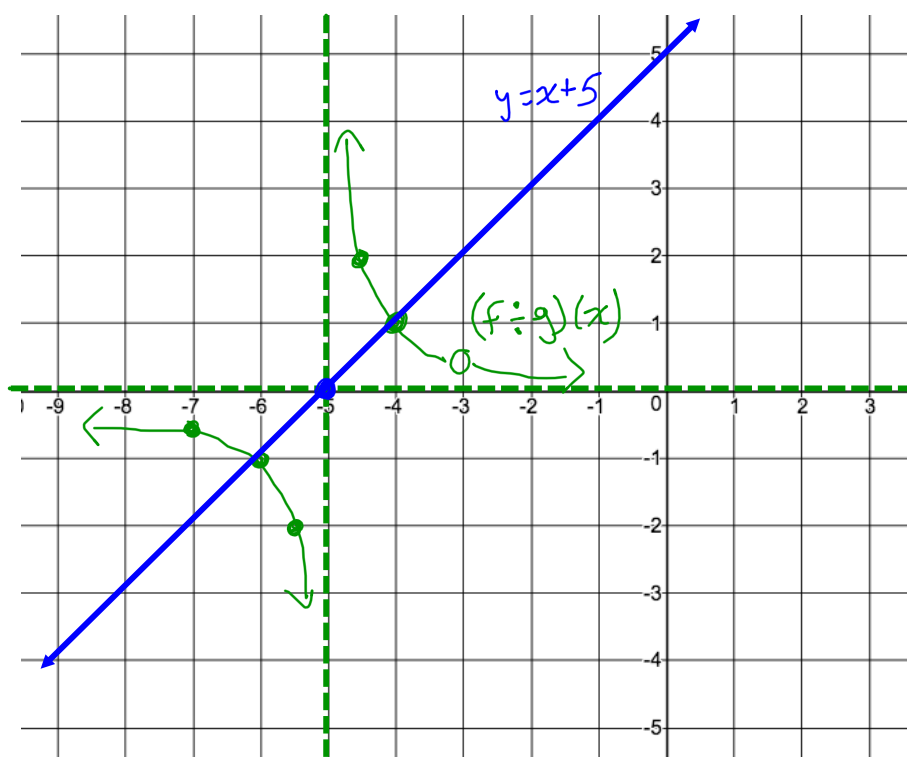
HA:  $y = 0$

Hole at  $x = -3$

**Note:** always a HA at  $y = 0$  when denominator is higher degree than numerator

| $y = x + 5$ |      |
|-------------|------|
| $x$         | $y$  |
| -7          | -2   |
| -6          | -1   |
| -5.5        | -0.5 |
| -5          | 0    |
| -4.5        | 0.5  |
| -4          | 1    |
| -3          | 2    |

| $y = \frac{1}{x+5}$ |               |
|---------------------|---------------|
| $x$                 | $\frac{1}{y}$ |
| -7                  | -0.5          |
| -6                  | -1            |
| -5.5                | -2            |
| -5                  | Und           |
| -4.5                | 2             |
| -4                  | 1             |
| -3                  | Und           |



c) State the domain and range of both functions

$$(f \times g)(x)$$

$$D: \{X \in \mathbb{R}\}$$

$$R: \{Y \in \mathbb{R}\}$$

$$(f \div g)(x)$$

$$D: \{X \in \mathbb{R} | x \neq -5, -3\}$$

$$R: \{Y \in \mathbb{R} | y \neq 0, 0.5\}$$